

Provision of Radiology Reports Simplified With Large Language Models to Patients With Cancer: Impact on Patient Satisfaction

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ABSTRACT

ACCOMPANYING CONTENT

PURPOSE To explore the perceived utility and effect of simplified radiology reports on oncology patients' knowledge and feasibility of large language models (LLMs) to generate such reports.

MATERIALS AND METHODS This study was approved by the Institute Ethics Committee. In phase I, five state-of-the-art LLMs (Generative Pre-Trained Transformer-4o [GPT-4o], Google Gemini, Claude Opus, Llama-3.1-8B, and Phi-3.5-mini) were tested to simplify 50 oncology computed tomography (CT) report impressions using five distinct prompts with each LLM. The outputs were evaluated quantitatively using readability indices. Five LLM-prompt combinations with best average readability scores were also assessed qualitatively, and the best LLM-prompt combination was selected. In phase II, 100 consecutive oncology patients were randomly assigned into two groups: original report (received original report impression) and simplified report (received LLM-generated simplified versions of their CT report impressions under the supervision of a radiologist). A questionnaire assessed the impact of these reports on patients' knowledge and perceived utility.

RESULTS In phase I, Claude Opus-Prompt 3 (explain to a 15-year-old) performed slightly better than other LLMs, although scores for GPT-4o, Gemini, Claude Opus, and Llama-3.1 were not significantly different ($P > .0033$ on Wilcoxon signed-rank test with Bonferroni correction). In phase II, simplified report group patients demonstrated significantly better knowledge of primary site and extent of their disease as well as showed significantly higher confidence and understanding of the report ($P < .05$ for all). Only three (of 50) simplified reports required corrections by the radiologist.

CONCLUSION Simplified radiology reports significantly enhanced patients' understanding and confidence in comprehending their medical condition. LLMs performed very well at this simplification task; therefore, they can be potentially used for this purpose, although there remains a need for human oversight.

 [Data Sharing Statement](#)
 [Data Supplement](#)

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INTRODUCTION

The ever increasing access to electronic health records enables patients to take part in their clinical decisions, thereby improving the overall quality of patient care.^{1,2} However, radiology reports, a pivotal component of clinical information, are often filled with complex medical terms and abbreviations as they are targeted at the referring clinicians, rather than the patients. This makes it challenging for the patients to comprehend this information and thus may result in anxiety and confusion, especially for those with abnormal radiology findings. In lieu of other reliable sources,

most of the patients end up depending on the treating physician to interpret and further explain the radiology reports to the patients, which in turn further increases the clinician workload.³

Large language models (LLMs) are advanced artificial intelligence (AI) models trained on vast amounts of text data, enabling them to understand, generate, and simplify human language. They use deep learning techniques, particularly transformer architectures, to analyze and predict sequences of words, making them useful for tasks such as summarization, translation, and natural language understanding.⁴ By

CONTEXT

Key Objective

Can simplified radiology reports generated using large language models (LLMs) positively affect the knowledge and perception of patients with cancer?

Knowledge Generated

LLMs show variable performance for simplification of reports with different prompts and it is essential to select the best model-specific prompt. Such simplified reports lead to significantly better knowledge levels, confidence, and understanding in patients about their disease, although there remains a need for human oversight.

Relevance (M. Behera)

This work aims to help patients being better informed about their disease and make knowledgeable treatment decisions.*

*Relevance section written by JCO Clinical Cancer Informatics Associate Editor Madhusmita Behera, PhD.

leveraging the capabilities of LLMs, the potential for transforming radiology reports into patient-friendly narratives becomes a tangible reality.⁵ This approach aligns with the global shift toward empowering patients in their health care decisions, recognizing that well-informed patients are more likely to engage actively in their treatment plans.^{6,7} Previous studies have explored the potential of LLMs for simplification of radiology reports and medical documents, but only from a clinician's perspective.^{8–12} To the best of our knowledge, there is no study addressing the actual implications of such AI-generated simplified versions of their reports on actual patient experience.

Thus, in this pilot study we aim to explore the impact of simplifying radiology reports into easily comprehensible narratives on patients' knowledge and perception, and feasibility of using LLMs to generate such reports.

MATERIALS AND METHODS

This was a prospective study approved by the Institute Ethics Committee. We performed the study in two phases.

Phase I—Testing of Different LLMs and Prompt Engineering for Report Simplification

In this initial phase, we tested the performance of five state-of-the-art LLMs—three closed-source models (Generative Pre-Trained Transformer-4o (GPT-4o), Google Gemini, Claude Opus) and two open-source models (Llama 3.1-8B and Phi-3.5-mini)—for the simplification of radiology report impressions. For this purpose, computed tomography (CT) reports for scans performed from January 2023 to June 2023 were collected from the Radiology Information System (RIS) in our tertiary-care oncology center. Of these, we selected 50 reports using stratified random sampling to ensure diversity in both the sites of primary cancers and

patient disease status. The patient demographics and cancer characteristics of this testing cohort are summarized in the Data Supplement (Table S1).

The impression sections of these reports were fed into the LLMs along with the prompt. Five distinct prompts were tested with each LLM to accomplish simplification of the medical jargon into layman's language. The complete prompts used for testing of the LLMs are detailed in the Data Supplement (Table S2). The readability of simplified reports generated for each LLM-prompt combination was quantitatively assessed using established readability scores: Gunning Fog Index, Flesch-Kincaid Grade Level, Automated Readability Index, Coleman-Liau Index, and an average of all four readability scores.¹³ These scores indicate the ease with which a text can be understood: lower scores suggest easier readability, whereas higher scores indicate increased complexity. For each LLM, the best performing prompt on quantitative evaluation was identified on the basis of the lowest average readability score, signaling a more accessible and easier-to-read output. The outputs from these five LLM-prompt combinations with the best average readability score were then qualitatively evaluated by a certified radiologist (8 years of experience in body imaging) for accuracy and completeness, and by a technical staff (5 years of experience in radiology-related research) for comprehensibility using a three-point Likert scale (Data Supplement, Table S3).

The LLM-prompt combination with the best performance on both quantitative and qualitative assessments was used in the next phase of the study.

Phase II—Providing Simplified Reports to the Patients and Patient-Directed Questionnaire

In the subsequent phase, we randomly provided the impression section of either the original radiology report or

their simplified versions to the patients and assessed their impact through a patient-directed questionnaire. The study was conducted in the oncology wing of a tertiary care facility. To keep the information provided to the patient succinct and avoid overwhelming them with less relevant findings, we used only the impression section of the reports, which routinely includes all the important information about the patient's tumor, including size and extent of the mass, metastatic involvement, and comparison with previous available imaging. Consecutive patients who underwent CT scans for baseline staging or response evaluation of cancer were included. The patients who were not able to read and understand English were excluded from the study. At our oncology center, all CT scans are requested by concerned oncologists after relevant clinical work-up and discussion with the patient. The CT reports are routinely provided to the patients in printed form the next day after the scan. This printed report is currently the only method for communicating scan results to patients, as the reports are not uploaded to an online patient portal and remain accessible solely within the RIS of our department. The patients were enrolled in the study when they visited the radiology department to collect their CT report, and the questionnaire-based interview was done at this time. Patients were explained about the nature and purpose of the study, and informed written consent was taken. A total of 100 patients were included in the study. The patient interview was done with an oncologist and a psychiatrist present on-site, with a provision for specialist consultation immediately after the interview, if needed.

Development and Validation of Questionnaire

The questionnaire was designed to assess the impact of providing the patients with their radiology report impressions. The brief study questionnaire was prepared by study authors on the basis of their expertise and review of relevant extant literature.^{14,15} Its face validity was established by modifying it as per the inputs received from experts, from various medical disciplines including Radiology (D.K.), Medical Oncology (A.B.), and Psychiatry (S.S.). These experts had clinical experience with oncology patients for 20, 15, and 7 years, respectively. Emphasis was given to using simple and understandable phrases while drafting questions, minimizing technical jargon. The questionnaire was pilot-tested with five patients with cancer to assess the suitability of items. On the basis of their feedback, ambiguous or unclear items were further refined and the questionnaire was finalized (Data Supplement).

Patient Interaction and Questionnaire-Based Assessment

The patients were randomly divided into two groups: original report and simplified report. These patients were presented with the impression section from their original report or the LLM-generated simplified impression, respectively. For the simplified report group, the impression section of the

patient's radiology report was fed into the selected LLM along with the best performing prompt (from phase I of the study) for simplification. For patients having multiple previous scans, only the recent-most CT report was used for this study as the comparison from the previous scans is itself routinely included within the impression section of the CT reports at our department. The LLM-generated simplified versions of the reports were first checked for any discrepancies by a supervising radiologist, who ensured that the simplified report accurately reflected the findings of the original report without altering the primary diagnostic content. Any required corrections were noted. The patients were then interviewed using the questionnaire after providing the generated reports. The questionnaire was administered to the patients by a radiology technical staff under the supervision of the radiologist. All patients read the radiology reports themselves.

In both the groups, patients were initially presented with the CT report impression (either original or simplified, according to the randomized group), followed by a set of questions aimed at gauging their understanding of the disease, as well as their confidence in interpreting the report. Time taken by the patients to read and process the information in the given radiology report was measured for both the groups. Aspects such as understandability of the report and any consequent psychological distress were also scrutinized. Any additional queries posed by the patients were also recorded. A flow diagram outlining the two phases of the study is shown in [Figure 1](#). The study workflow is summarized in [Figure 2](#). Two examples of simplified reports generated using the LLM are shown in the Data Supplement (Fig S1).

Statistical Analysis

All the statistical analyses were performed using the R programming language (R Core Team, 2023). In phase I, to assess the differences in quantitative readability scores among simplified versions of radiology reports, nonparametric Friedman's test was used (due to non-normal data distribution as indicated by the initial Shapiro-Wilk test; $P < .05$). Subsequently, post hoc pairwise comparisons were conducted using the Wilcoxon signed-rank test with Bonferroni correction to control for multiple comparisons (adjusted significance level of $P < .0033$). Descriptive analysis of the qualitative scores for LLM outputs was performed. In the questionnaire phase of the study, the responses obtained from the participants were analyzed using descriptive statistics. The comparative analysis of Likert scores in the two groups (conventional report v simplified report) was performed using appropriate nonparametric tests (chi-square test for nominal data and Mann-Whitney U test for ordinal data).

Ethics Approval and Consent to Participate

Ethics approval was obtained from the Institutional Review Board. Consent was obtained from each patient before recruitment in the study.

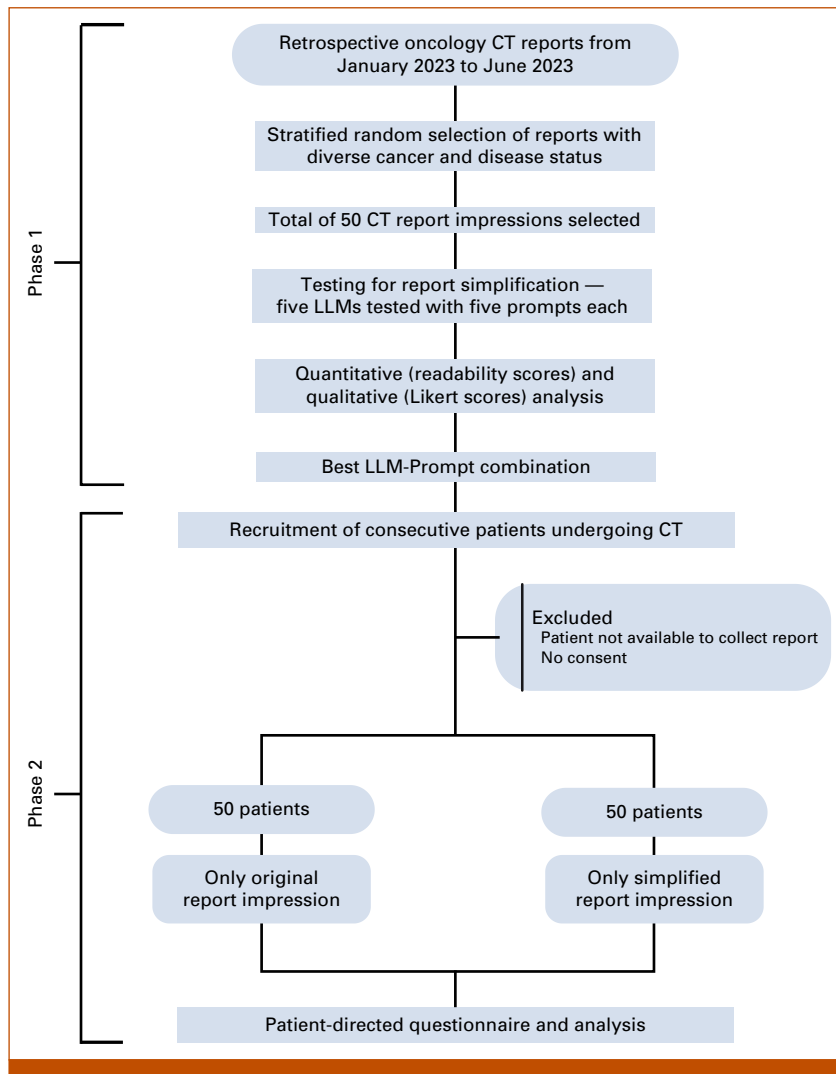


FIG 1. Flow diagram to show the two phases of the study. CT, computed tomography; LLM, large language model.

RESULTS

Comparison of Performance of Different LLM-Prompt Combinations for Simplification of Reports

The Data Supplement (Table S4) shows the mean values of various readability indices for the original reports and the simplified reports generated using the five LLMs, each with five different prompts. Of these, the best performing prompt (with lowest average readability score) for each LLM was used for further analysis. These five LLM-prompt combinations were as follows: GPT-4o-Prompt 3, Gemini-Prompt 3, Claude-Prompt 3, Llama-Prompt 5, and Phi-Prompt 5. Table 1 summarizes the average readability scores and their statistical comparisons among original reports and outputs from the best performing LLM-prompt combinations. The detailed pairwise comparisons and significance of readability scores for these reports and their average Likert scores are shown in the Data Supplement (Tables S5 and S6), respectively.

The analyses demonstrated that all LLM outputs simplified the reports to a significant level compared with the complex language in the original reports (all $P < .001$). Significant differences were also observed among the LLMs. Specifically, Phi's output differed significantly from those of Llama ($Z = -3.142$; $P = .002$), Claude ($Z = -3.556$; $P < .001$), and Gemini ($Z = -3.567$; $P < .001$), indicating variability in simplification performance among the models. No significant differences were found between GPT-4o, Llama, Claude, and Gemini, suggesting comparable performance among these LLMs in terms of readability enhancement. Notably, although the mean readability difference between GPT-4 and Claude was larger than that between Phi and Llama, the higher standard deviation in GPT-4 scores (3.26; Table 1) introduced greater variability, which reduced the statistical power to detect a significant difference with Claude ($P = .310$; $Z = -1.022$). In contrast, the lower standard deviations of Phi and Llama (2.12 and 1.73, respectively) indicate more consistent scores, resulting in a statistically significant difference between

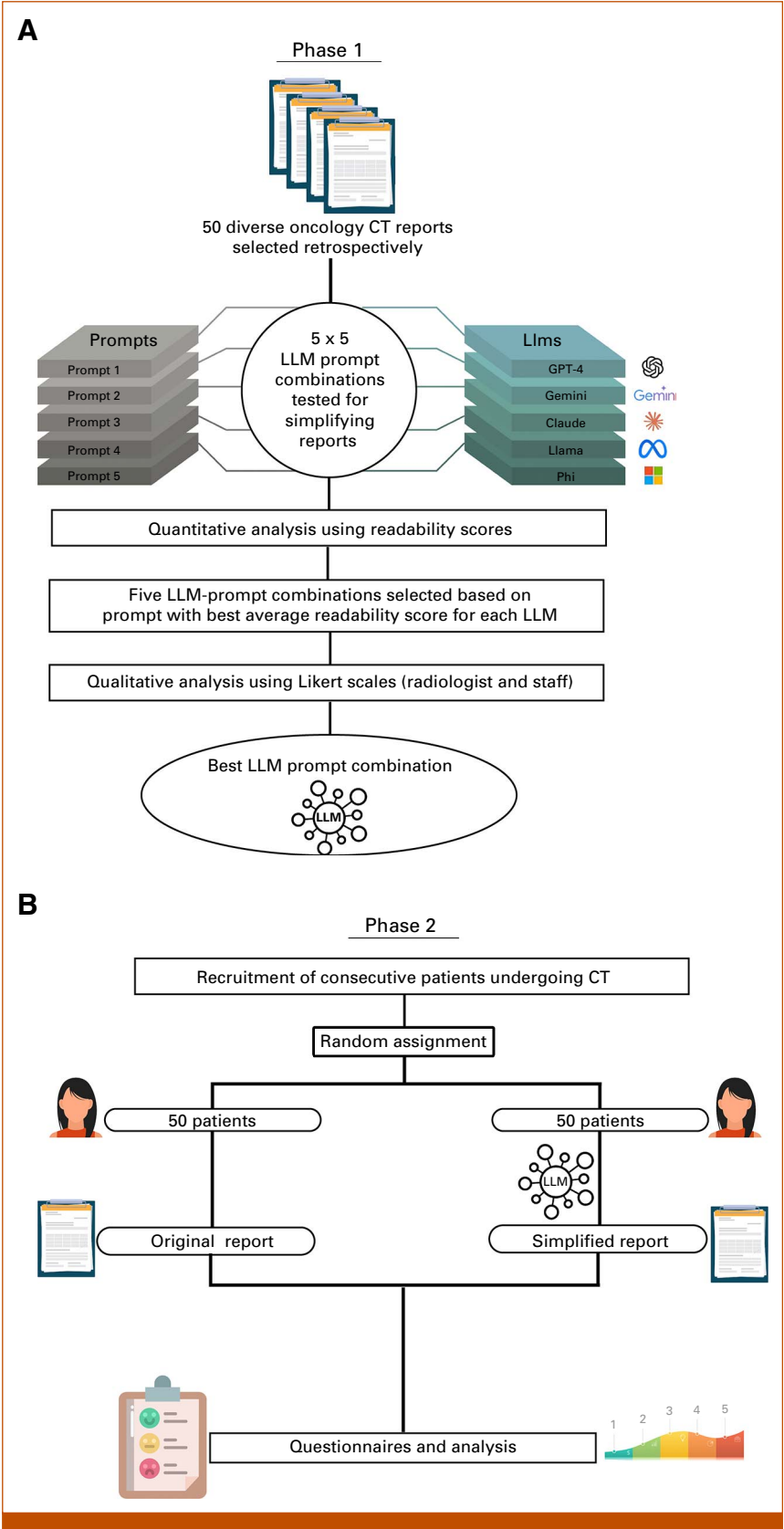


FIG 2. Workflow summarizing the (A) phase I and (B) phase II of the study. CT, computed tomography; GPT-4, Generative Pre-Trained Transformer-4; LLM, large language model.

TABLE 1. Average Readability Scores and Statistical Comparisons Among Original Reports and Outputs From Best Performing LLM-Prompt Combinations

Report/LLM-Prompt Combination	Mean Readability Score	Standard Deviation	Significantly Different from LLMs
Original report	15.01	4.05	All LLMs ($P < .001$)
GPT-4-prompt 3	8.36	3.26	None
Gemini-prompt 3	7.03	1.93	Phi ($P < .001$)
Claude-prompt 3	6.96	1.32	Phi ($P < .001$)
Llama-prompt 5	7.65	1.73	Phi ($P = .002$)
Phi-prompt 5	8.87	2.12	Llama, Claude, Gemini ($P < .0033$)

Abbreviations: GPT-4, Generative Pre-Trained Transformer-4; LLM, large language model.

these models ($P = .002$; $Z = -3.142$). The applied Wilcoxon signed-rank test with Bonferroni correction appropriately accounts for both the mean differences and the consistency of the scores in determining statistical significance for pairwise comparisons.

Since the combination of Claude Opus with prompt 3 (simplify for a 15-year-old) had a slightly better performance on both the qualitative and quantitative readability scores compared with other LLM-prompt combinations, it was chosen to be used in the questionnaire phase of the study.

Assessment of the Patient Responses Recorded Through the Questionnaire

The demographics, cancer characteristics, and educational levels of the interviewed patients in the two groups (original and simplified reports) are detailed in Table 2. The mean age of study participants was 45.7 years (M:F = 28:22) in the original report group and 40.1 years (M:F = 30:20) in simplified report group.

The patients' knowledge levels about the primary cancer site and extent of their disease are shown in the Data Supplement (Fig S2). The chi-square test of independence showed a significant association between the type of report (original v simplified) and patients' knowledge for both the primary site of cancer ($P = .0016$) and the extent of disease ($P < .0001$). These results suggest that the simplified reports significantly improved patients' knowledge of their disease compared with the original reports. The comparative analysis of the patients' knowledge levels in the two groups is shown in Table 3. Of note, the patients with only up to primary education level, those having baseline imaging, and those diagnosed with abdomino-pelvic malignancy (not discretely evident to the patient) had wrong knowledge regarding the primary site ($n = 5/7$, $n = 5/12$, $n = 5/25$, respectively) and extent ($n = 6/7$, $n = 6/12$, $n = 17/25$, respectively) of their disease in the original report group, but patients with these factors had either a partially correct or correct knowledge about the primary site and a much lower proportion had wrong information about the disease extent ($n = 2/6$, $n = 2/13$, $2/22$) on reading the simplified reports (refer to the Data

Supplement, Tables S7–S9 for detailed descriptive analysis of patient responses stratified according to these factors).

The patients' confidence and understanding of the reports, as recorded using a five-point Likert scale, are shown in the Data Supplement (Fig S3). The Mann-Whitney U test revealed a significant difference in patients' confidence ($P = .007$) and understanding of reports ($P < .0001$) between the original and simplified report groups. These findings indicate that patients in the simplified report group demonstrated significantly higher confidence and better understanding compared with the original report group. The comparative analysis is shown in Table 4. Considering the lowest two Likert scores, patients with lower education levels (up to primary level) felt underconfident ($n = 3/7$) and failed to understand ($n = 7/7$) the original reports, but those in the simplified report group had better scores, none with the lowest two Likert scores (refer to the Data Supplement, Tables S7–S9 for detailed descriptive analysis of patient responses stratified according to these factors).

The mean time taken by the patients to read and interpret the given reports was 71.4 seconds (range, 25–110 seconds) for the original report group and 65.5 seconds (range, 20–134 seconds) for the simplified report group. In both the groups, few patients ($n = 5/50$ in the original report group and $n = 6/50$ in the simplified report group) reported increased psychological distress after reading their reports. These patients either had progressive disease or only a partial response on the current scan. These patients were then provided psychiatry and medical oncology consultation on a priority basis on the same day. Patients who were not able to answer the questions about their disease primary and extent correctly as well as those who were not confident about their understanding were further explained about their report by the technical staff under the supervision of the radiologist. Twenty-two patients ($n = 13/50$ in the original report group and $n = 9/50$ in the simplified report group) raised further queries regarding future treatment and prognosis, and they were advised to consult with their treating physician for further discussion.

Notably, three (of 50) of the LLM-generated reports required corrections by the supervising radiologist. These mistakes

TABLE 2. Demographics and Cancer Characteristics of the Interviewed Patients

Characteristic	Original Report Group (n = 50)	Simplified Report Group (n = 50)
Patient demographics		
Age, years, mean (range)	45.7 (25-78)	40.1 (22-75)
Gender, No. (%)		
Male	28 (56)	30 (60)
Female	22 (44)	20 (40)
Education levels, No. (%)		
Primary education	7 (14)	6 (12)
Secondary education	16 (32)	18 (36)
Higher secondary education	10 (20)	13 (26)
Undergraduate degree	12 (24)	10 (20)
Postgraduate degree	5 (10)	3 (6)
Primary cancer sites and disease status, No. (%)		
Head and neck		
Upper aero-digestive tract	9 (18)	5 (10)
Salivary glands	0	2 (4)
Chest		
Lung	6 (12)	4 (8)
Mediastinal lymphoma	3 (6)	4 (8)
Breast	1 (2)	4 (8)
Abdomen		
Bowel	10 (20)	6 (12)
Gall bladder	2 (4)	0
Stomach	1 (2)	3 (6)
Pancreas	3 (6)	1 (2)
Liver	1 (2)	0
Kidney	3 (6)	2 (4)
Pelvis		
Ovary	1 (2)	2 (4)
Cervix	4 (8)	6 (12)
Urinary bladder	0	1 (2)
Rectum	0	0
Testis	0	1 (2)
Bone tumors and extremities		
Soft-tissue sarcoma	2 (4)	4 (8)
Osteosarcoma	4 (8)	5 (10)
Status of disease compared with previous scan (n = 50)		
Baseline	12 (24)	13 (26)
Complete response	3 (6)	4 (8)
Partial response	12 (24)	18 (36)
Stable disease	17 (34)	11 (22)
Progressive disease	6 (12)	4 (8)

were a misinterpretation of overlapping medical terms and medical abbreviations—specifically, post-radiotherapy changes—misinterpreted as changes in the right side (instead of radiotherapy changes), retro-molar trigone misinterpreted as masseteric area (instead of retro-molar trigone), and vault misinterpreted as upper part of the abdominal cavity (instead of posthysterectomy vaginal vault).

DISCUSSION

This study explored the impact of using simplified radiology reports in layman language in a tertiary oncology setup, finding that such reports enhanced patients' knowledge and comprehension, indicating a significant benefit in conveying complex medical information. Using LLMs provided an efficient automated method to generate such reports.

TABLE 3. Knowledge of Patients About Their Disease (site of primary and disease extent) in the Two Groups (original v simplified report)

Study Variable	Patient Group	Does Not Know/Wrong Knowledge, No. (%)	Partially Correct Knowledge, No. (%)	Correct Knowledge, No. (%)	Test Statistic— χ^2 (degree of Freedom) ^a	P
Site of primary	Original report group (n = 50)	5 (10)	12 (24)	33 (66)	12.85 (2)	.0016
	Simplified report group (n = 50)	0	3 (6)	47 (94)		
Extent of disease	Original report group (n = 50)	22 (44)	9 (18)	19 (38)	22.35 (2)	<.0001
	Simplified report group (n = 50)	2 (4)	12 (24)	36 (72)		

^aChi-square test of independence.

TABLE 4. Patients' Confidence About Their Disease Knowledge and Understandability of Reports in the Two Groups (original v simplified report)

Study Variable	Patient Group	1, ^a No. (%)	2, ^a No. (%)	3, ^a No. (%)	4, ^a No. (%)	5, ^a No. (%)	Test Statistic (<i>U</i> -statistic) ^b	P
Patients' confidence	Original report group (n = 50)	4 (8)	6 (12)	9 (18)	10 (20)	21 (42)	886.5	.007
	Simplified report group (n = 50)	0	1 (2)	2 (4)	18 (36)	29 (58)		
Patients' understanding of reports	Original report group (n = 50)	33 (66)	7 (14)	7 (14)	2 (4)	1 (2)	78.5	<.0001
	Simplified report group (n = 50)	0	1 (2)	3 (6)	15 (30)	31 (62)		

^aFive-point Likert scale (1: not at all confident/very difficult to understand, 5: extremely confident/easily understandable).

^bMann-Whitney *U* test.

Within a year of release of initial GPT models, the literature is replete with various potential LLM applications in radiology.^{16,17} Although previous studies demonstrate efficient radiology report simplification by LLMs like GPT-3.5, GPT-4, Bard, and Bing, none have compared open- and closed-source models.^{5,8-12} In our study, we found that the open-source Llama-3.1 model had a comparable performance than the tested closed-source models. Our study also highlights the need for model-specific prompts, with the Claude Opus-prompt 3 (simplify for a 15-year-old) combination slightly outperforming others.

Notably, the available literature on using LLMs to simplify radiology reports suffers from various drawbacks. Existing studies assess simplified reports from the perspective of radiologists or medical experts, often using retrospective or limited-scope reports, offering limited real-life clinical utility for the patients.⁸⁻¹² Our study improves on this by testing LLM-simplified reports with actual patients, overseen by a radiologist. We demonstrate that such simplified reports lead to significantly better knowledge levels, confidence, and understanding in patients about their disease. Patients with cancer with some unique factors like lower education levels, presence of a malignancy that is not easily appreciable to the patient (like abdomino-pelvic region), patients undergoing first baseline scan for the disease, or those with only partial response are particularly likely to benefit from such simplified versions of their reports. Given the high workload for oncologists, especially in countries with large populations and varying patient literacy levels, the physician-patient relationship is significantly stressed leading to limited communication of their disease status to the patients in many instances.¹⁸ In this regard, our study represents a significant leap forward in using the LLMs for positively affecting the actual patient experience. Also, the instances where the simplifications required corrections by a radiologist, although minimal, are a crucial reminder of the technology's current limitations.

Our study also examined risks associated with LLM use such as data privacy, potential misinterpretation of medical terms, the need for further explanation, and the possibility of increased patient anxiety for reports indicating disease progression. Although we used a closed-source LLM in the patient phase of our pilot study (owing to a better performance in the prompt engineering phase), using a locally run

open-source LLM (like Llama in this study) can completely obviate the need to share patient data with a third-party server. Also, the performance and consistency of such LLMs can be improved even further by established techniques including retrieval-augmented generation and fine-tuning.¹⁹ These techniques need to be further explored to establish whether they can refine the LLM performance to such a level as to obviate the need for further review or explanation by the radiologist, and at the same time, to phrase the report language in a way that is less distressing for the patients, especially for reports bearing grave consequences for them.

Our study has certain limitations. The relatively small sample size and the specific context of oncology patients at a tertiary-care center may limit generalizability. Excluding patients unable to read and understand English restricts applicability to English-speaking patients, as we focused on report simplification alone without combining translation. In the future, we aim to explore LLM translation capabilities to enhance generalizability across diverse populations. Subjective Likert scales and potential biases in patient feedback could skew some results. We provided detailed stratified analyses on the basis of various patient characteristics to address potential confounding factors; however, we acknowledge that future research with pre-and-post assessments of imaging-specific knowledge may further isolate the impact of report simplification. Although we focused our study on the impression section, patients usually access their full reports and may have concerns about other sections, necessitating a broader approach. Workflow variability, electronic health records integration, and asynchronous report delivery also need to be addressed to generalize this system effectively.

In conclusion, the introduction of LLM-generated simplified radiology reports significantly enhances patient understanding, confidence, and satisfaction in comprehending their medical conditions. Although the need for human oversight remains, the benefits of such technological integration in enhancing patient communication are undeniable. This study paves the way for further exploration and refinement of AI tools in patient-centered healthcare, advocating for a balance between technological innovation and expert human oversight.

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DATA SHARING STATEMENT

Data generated or analyzed during the study are available from the corresponding author by request.

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Accountable for all aspects of the work: All authors

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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Open Payments is a public database containing information reported by companies about payments made to US-licensed physicians ([Open Payments](http://OpenPayments.org)).

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REFERENCES

1. Kaba R, Sooriakumaran P: The evolution of the doctor-patient relationship. *Int J Surg* 5:57-65, 2007
2. Anyidoho PA, Verschraegen CF, Markham MJ, et al: Impact of the immediate release of clinical information rules on health care delivery to patients with cancer. *JCO Oncol Pract* 19:e706-e713, 2023
3. Mehan WA, Gee MS, Egan N, et al: Immediate radiology report access: A burden to the ordering provider. *Curr Probl Diagn Radiol* 51:712-716, 2022
4. Thirunavukarasu AJ, Ting DSJ, Elangovan K, et al: Large language models in medicine. *Nat Med* 29:1930-1940, 2023
5. Jeblick K, Schachtner B, Dext J, et al: ChatGPT makes medicine easy to swallow: An exploratory case study on simplified radiology reports. *Eur Radiol* 34:2817-2825, 2024
6. Russo G, Moretta Tartaglione A, Cavacece Y: Empowering patients to co-create a sustainable healthcare value. *Sustainability* 11:1315, 2019
7. Bailo L, Guidi P, Vergani L, et al: The patient perspective: Investigating patient empowerment enablers and barriers within the oncological care process. *Ecancermedicalscience* 13:912, 2019
8. Lyu Q, Tan J, Zapadka ME, et al: Translating radiology reports into plain language using ChatGPT and GPT-4 with prompt learning: Results, limitations, and potential. *Vis Comput Ind Biomed Art* 6:9, 2023
9. Doshi R, Amin K, Khosla P, et al: Utilizing large language models to simplify radiology reports: A comparative analysis of ChatGPT-3.5, ChatGPT-4.0, Google Bard, and Microsoft Bing. *medRxiv* 10.1101/2023.06.04.23290786
10. Amin KS, Davis MA, Doshi R, et al: Accuracy of ChatGPT, Google Bard, and Microsoft Bing for simplifying radiology reports. *Radiology* 309:e232561, 2023
11. Sarangi PK, Lumbani A, Swarup MS, et al: Assessing ChatGPT's proficiency in simplifying radiological reports for healthcare professionals and patients. *Cureus* 15:e50881, 2023
12. Doshi R, Amin KS, Khosla P, et al: Quantitative evaluation of large language models to streamline radiology report impressions: A multimodal retrospective analysis. *Radiology* 310:e231593, 2024
13. Chen W, Durkin C, Huang Y, et al: Simplified readability metric drives improvement of radiology reports: An experiment on ultrasound reports at a pediatric hospital. *J Digit Imaging* 30:710-717, 2017
14. Andrade C, Menon V, Ameen S, et al: Designing and conducting knowledge, attitude, and practice surveys in psychiatry: Practical guidance. *Indian J Psychol Med* 42:478-481, 2020
15. Farooq R: How to design and frame a questionnaire, in John Jeyasekar J, Saravanan P (eds): *Innovations in Measuring and Evaluating Scientific Information*. Hershey, PA, IGI Global, 2018, pp 50-60
16. Srivastav S, Chandrakar R, Gupta S, et al: ChatGPT in radiology: The advantages and limitations of artificial intelligence for medical imaging diagnosis. *Cureus* 15:e41435, 2023
17. Elkassem AA, Smith AD: Potential use cases for ChatGPT in radiology reporting. *AJR Am J Roentgenol* 221:373-376, 2023
18. Fundytus A, Sullivan R, Vanderpuye V, et al: Delivery of global cancer care: An international study of medical oncology workload. *JCO Glob Oncol* 10.1200/JGO.17.00126
19. Zhang G, Jin Q, Zhou Y, et al: Closing the gap between open source and commercial large language models for medical evidence summarization. *NPJ Digit Med* 7:239, 2024